Lab 3: Component Modelling & Architecture

Incident Escalation & On-Call Rotation Engine

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Architecture selection and alternatives

We chose Layered Architecture for the Incident Escalation & On-Call Rotation Engine. The presentation/integration layer contains the Operations Portal and Alert & Callback Gateway; the business layer contains the Escalation & Acknowledgement Manager, Rotation & Override Manager, Notification Adapter, and a Post-mortem Manager; the data layer contains the data acess Repository and Incident Database. External messaging providers are accessed through the adapter.

Layered architecture provides clear responsibilities and local calls, but its application tiers cannot scale independently. Microservices offer independent scaling and fault isolation for notification channels, but add network latency, deployment overhead, and consistency challenges between acknowledgements and timers. Client-server simplifies central control and consistent records, but a single server can become a bottleneck or failure point. A layered application is the best fit for the current lab scope; redundancy remains a deployment requirement.

Two scenario-specific reasons

1. Reliable incident lifecycle: one business module owns routing, acknowledgement, and the five-minute escalation deadline. Transactional incident updates and persisted deadlines let workers recheck acknowledgement before escalating, suppress duplicate callbacks, and recover pending work after a restart.

2. Separation of changing rules: shift assignments, active/upcoming overrides, and replacement-engineer lookup belong to the Rotation & Override Manager. Messaging and post-mortem documentation have separate interfaces, so changing an SMS provider or adding root-cause/action-item fields does not change routing logic. These modules cover all five Lab 2 epics.

Security advantage

Enforce role checks in the business layer: engineers acknowledge permitted incidents, while commanders/admins manage overrides and schedules. Validate signed monitoring/provider callbacks at the gateway and use HTTPS at external boundaries. Only the repository accesses the database with least-privilege credentials and parameterized queries; record acknowledgements and overrides in an audit trail.

Performance benefit

Local business calls avoid inter-service network hops on the alert path. Indexed on-call lookups and asynchronous notification workers keep slow provider responses outside ingestion processing. Persist notification work before dispatch and invalidate schedule caches on edits/overrides. Target dispatch initiation within three seconds of ingestion; load tests must verify this target. Unacknowledged incidents escalate after five minutes.